

**PRESIDENT'S OFFICE REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
MOROGORO REGION MOCK EXAM
FORM TWO MOCK EXAMINATION**

031 PHYSICS MARKING SCHEME

Time: 2:30 Hours

Year; 2026

SECTION A (15 Marks)

1.

i	ii	iii	iv	v	vi	vii	viii	ix	x
A	A	C	C	A	E	D	D	B	A

2.

GEPAM science hub

(@1%, Total = 10 marks)

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LIST A	i	ii	iii	iv	v
LIST B	C	A	E	D	H

(5 Marks)

SECTION B (70 Marks)

3. (a) (i) Three conditions to be satisfied before a body can float.
- ♣ The overall density of a floating body should be lower than the density of a liquid in which it is floating
 - ♣ The body should be large and hollow
 - ♣ Upthrust on a body should be equal to the weight of a body in the air.
- (01@ =03 marks)**

- (ii) The three factors affecting the upthrust of a fluid are
- ♥ the volume of the immersed object
 - ♥ The density of the fluid in which the object is immersed
 - The acceleration due to gravity **(1 @ = 03 marks)**

- (b) The relative density of the liquid.

Weight of a substance in air, $W_1 = 15\text{N}$

Apparent weight in water, $W_2 = 11\text{N}$

Apparent weight in a Liquid, = 9N

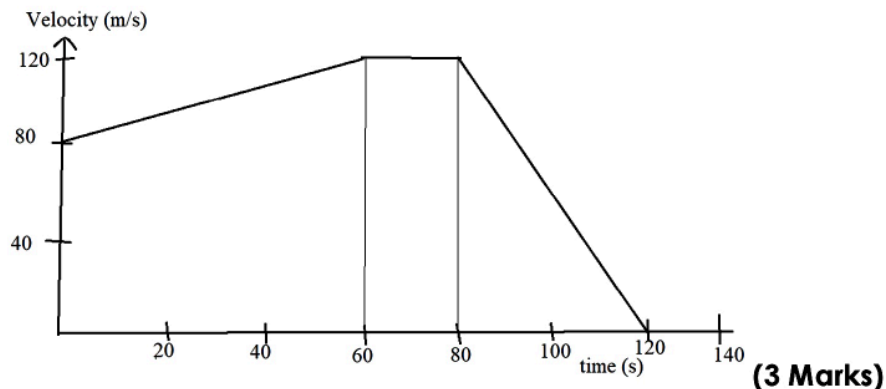
$$\text{Relative density} = \frac{\text{Upthrust in a liquid}}{\text{Upthrust in water}} \quad \mathbf{01 \text{ mark}}$$

$$\text{Relative density} = \frac{W_1 - W_3}{W_1 - W_2} \quad \mathbf{01 \text{ mark}}$$

$$\text{Relative density} = \frac{15 - 9}{15 - 11} \quad \mathbf{01 \text{ mark}}$$

$$\therefore \text{Relative density} = 1.5 \quad \mathbf{(01 \text{ marks})}$$

4. The velocity - time graph for the journey. Hence from the graph calculate;



- (i) The deceleration of the car

$$\text{Acceleration, } a = \frac{v-u}{t}$$

$$a = \frac{0-120}{40} = -3\text{m/s}^2$$

$\therefore$ The deceleration = 3m/s^2 or acceleration = -3m/s^2 **(3 marks)**

- (ii) The total distance travelled by the car

Total distance travelled = Total area under the graph
= Area of trapezium + Area of a rectangle + Area of a triangle

$$\frac{1}{2}(a+b)h + lh + \frac{1}{2}ab$$

$$\frac{1}{2}(80+120)60 + 120 \times 20 + \frac{1}{2} \times 120 \times 40$$

$\therefore$ The total distance travelled = 10800m **(4 marks)**

5. (a) (i) Static electricity can be converted to current electricity by touching the area of high concentration of charge, or a negative terminal of a battery by using a wire. **(3 marks)**

- (ii) – Quantity of charge on an object.
– Distance between charged bodies. **(2 marks)**

(b)

$$\frac{1}{CT} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \quad \mathbf{01 \text{ mark}}$$

$$\frac{1}{CT} = \frac{1}{2} + \frac{1}{3} + \frac{1}{5} \quad \mathbf{01 \text{ mark}}$$

$$\frac{1}{CT} = \frac{31}{30} \quad \mathbf{01 \text{ mark}}$$

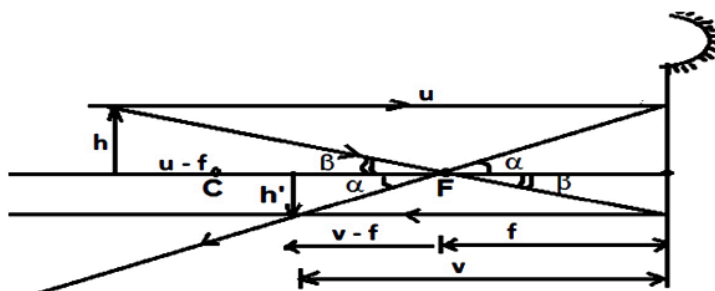
$$CT = 30/31$$

$$Q = CV \quad \mathbf{01 \text{ mark}}$$

$$Q = 30/31 \times 10 = 0.9677 \times 10$$

$\therefore$ Quantity of charge = 9.677C **(01 marks)**

6. (a) i.



(03 Marks)

ii. conditions for the total internal reflection of light **02 marks**

- The angle of incident should be greater than critical angle
- Light must travel from denser media toward less dense media

(b)

Focal length, $f = 12\text{cm}$, $m = 4$, $V = ?$

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\text{But } v = 4u$$

$$\text{Then; } \frac{1}{f} = \frac{1}{u} + \frac{1}{4u}$$

$$\frac{1}{f} = \frac{4+1}{4u} \text{ But } f = 12 \text{ cm}$$

$$\frac{1}{12} = \frac{5}{4u}$$

$$\therefore u = 15\text{cm}$$

$$\text{Since } v = 4u$$

$$\text{Then } v = 4 \times 15$$

$$\therefore \text{The image distance} = 60\text{cm from the lens}$$

The image is real and inverted, also magnified. **(05Marks)**

7. (a) (i) Osmosis is the movement of solvent molecules from the solution of low concentration to the solution of high concentration through a semipermeable membrane, while diffusion is the movement of particles from the higher concentration to the region of lower concentration. **(02 marks)**

(ii) The two factors affecting the capillarity of a capillary tube are;

- Diameter of the tube or opening
- Nature of the liquid **(02marks)**

(b)

$$l_1 = 20\text{cm}, l_2 = 25\text{cm } \Delta l = 25\text{cm} - 20\text{cm}$$

$$\therefore \Delta l = 5\text{cm}$$

$$\text{Percentage change in length} = \frac{\Delta l}{l_1} \times 100\%$$

$$\text{Percentage change in length} = \frac{5}{20} \times 100\%$$

$$\therefore \text{Percentage change in length} = 25\% \quad \textbf{(06marks)}$$

8. (a) This is due to the difference in atmospheric pressure. The pressure of the atmosphere is low at the mountain than at the foot of the mountain.

(02 marks)

(b)

- (i) the maximum potential energy of the bob

$$\text{Potential energy} = mgh$$

$$PE = 0.12 \times 10 \times 0.02$$

$$\therefore \text{Potential energy} = 0.024\text{J} \quad \textbf{(03 marks)}$$

- (ii) the maximum speed of the bob

$$\text{Kinetic energy} = \frac{1}{2}mv^2$$

$$KE = \frac{1}{2} \times 0.12 \times v^2$$

$$\therefore KE = 0.6V^2 \quad \textbf{(01 marks)}$$

But the maximum PE at max. displacement = maximum KE at equilibrium

$$\therefore 0.024 = 0.6V^2$$

$$V^2 = 0.04$$

$$\therefore \text{The maximum speed of the bob} = 0.2\text{m/s} \quad \textbf{(01 marks)}$$

- (iii) The periodic time of a bob.

$$T = 2\pi\sqrt{l/g}$$

$$T = 2\pi\sqrt{0.8/10}$$

$$\therefore \text{The periodic time of the bob} = 1.77\text{s} \quad \textbf{(03 Marks)}$$

9. (a)

- (i) The angle of declination is the angle between the magnetic north and geographic north, or is the angle between the magnetic meridian and geographic meridian.

(2 Marks)

- (ii) A magnetic pole is the end of a magnet corresponding to the north pole or south pole of the earth's magnet.

(2 Marks)

- (iii) Applications of magnets

- Used in speakers and microphones
- Used in magnetic recording media
- Used in transformers
- Used in electrical generators
- Tv and computer monitors
- Electromagnets
- Used in credit, debit and ATM cards

(any four points 00 $\frac{1}{2}$ @ = 2 marks)

(b) (i) We can test the polarity of a magnet by suspending it freely, then the north seeking pole is the south pole and the south seeking pole is the north pole. Also, by bringing a known pole near the unknown pole, if attraction occurs then poles are the same, but if repulsion occurs then the poles are unlike. **(02 Marks)**

(ii) Attraction is not a sure test for polarity because not only unlike poles of a magnet can be attracted by a magnet but also a magnetic material can be attracted. **(02 Marks)**

SECTION C (15 Marks)

10. (a) Draw the symbols of the following electric components **(01 mark @=4 marks)**

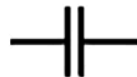
(i) Key



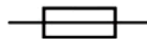
(ii) variable resistor



(iii) capacitor



(iv) fuse



(b)

(i) Switches K_1 and K_2 are closed

When switch K_1 and K_2 are closed current will pass through $5\ \Omega$ and $3\ \Omega$ resistor

Total resistance in series = $3\ \Omega + 5\ \Omega$ **01 Mark**

$$R_T = 8\ \Omega$$

$$\text{pd} = 2\ \text{V}$$

Therefore;

$$I = \frac{V}{R_T} \quad \mathbf{01\ mark}$$

$$I = \frac{2}{8}$$

$$I = 0.25\ \text{A} \quad \mathbf{01\ mark}$$

Therefore the total current flowing through the circuit is 0.25 A **01 mark**

(iii) Switch K_1 is open while switch K_2 is closed

When switch K_1 is open and K_2 is closed, then the current will not flow in the circuit at all

$\therefore I = 0A$ (02 marks)

(c) i. Joules law states that " The heat generated in a given time is directly proportional to the resistance of a conductor and the square of current " **02 marks**

ii. Ohms law " The current flowing in the conductor is directly proportional to the p.d across its end provided that temperature and other physical factors are constant" **02 marks**